

Soroush Omidvarkehrani

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Summary

Data scientist and AI researcher with a Ph.D. in Computer Science, building LLM and data-intensive systems in both production and research settings. Designed and shipped an enterprise RAG system for Canadian telecom operators, covering document onboarding, PII redaction, hybrid retrieval, and the evaluation methodology behind it. Author of papers at WWW, ADBIS, and TaDA@VLDB on LLM-based data wrangling (imputation, entity resolution, error detection, transformation), plus earlier work on large-scale streaming anomaly detection. Comfortable across the stack: retrieval architecture, LLM evaluation, large-scale ETL, and deployment.

Education

2022–2026 **Ph.D. in Computer Science**, University of Alberta, Canada (Expected Nov. 2026), GPA: 4

2017–2021 **M.Sc. in Computer Engineering**, Ferdowsi University of Mashhad, Iran

2013–2017 **B.Sc. in Computer Engineering**, Ferdowsi University of Mashhad, Iran

Experience

2025–2026 Data Scientist at Data on Motion Inc., Toronto, Canada

- Designed the architecture of an enterprise RAG system end to end: document onboarding, indexing, retrieval, and generation
- Defined the evaluation methodology as the basis for design decisions and to detect quality degradation as models and data evolved
- Built the document onboarding pipeline: layout-aware parsing of heterogeneous formats, structure-preserving chunking, PII redaction, and metadata enrichment to support filtered and scoped retrieval
- Implemented hybrid semantic/keyword retrieval with reranking over Milvus / Zilliz Cloud

2022–2026 AI Researcher at Data & Language Intelligence (DaLI) Lab, University of Alberta, Canada

- Built LLM-based systems for data wrangling: imputation, entity resolution, error and anomaly detection, and cell transformation
- Developed example-selection methods that pick diverse and discriminative in-context demonstrations, raising task accuracy while cutting prompt size against full-context baselines
- Designed sampling-based algorithms for discovering table transformations, reducing candidate search cost by orders of magnitude
- Benchmarked open-weight and API LLMs on accuracy, latency, and cost; ran large-scale experiments on HPC/SLURM clusters
- Teaching assistant for File and Database Management (4 years)

2019–2022 Project Manager at IP-PBX Type Approval Lab, Mashhad, Iran

- Led a research team delivering an industry-funded electricity-theft and illegal cryptocurrency-mining detection system for Khorasan Electrical Distribution Company; outcomes published in *Electric Power Systems Research*
- Architected a Spark-based streaming framework for near real-time anomaly detection over large-scale AMI smart-meter data, with a tunable trade-off between detection rate and false alarms
- Planned project scope and timelines, reported progress to the industry, and supervised a team of students across implementation and evaluation

Honors

2025, 2026 Graduate Student Teaching Award, University of Alberta

2024–2026 Alberta Innovates Graduate Student Scholarship (**AIGSS**), University of Alberta

2024 Flight PS752 Commemorative Scholarship

2023 Alberta Graduate Excellence Scholarship (**AGES**), University of Alberta

2022 Doctoral Recruitment Scholarship, University of Alberta

2021 **Best M.Sc. Thesis Award** at the 5th International Conference on Internet of Things and Applications

2021 **3rd place** in the Novel Ideas for Facing the Increase in Power Consumption and the Challenge of Power Outage event, Iran's National Elites Foundation

2020 **3rd place** among M.Sc. Computer Engineering students at Ferdowsi University of Mashhad

2019 **Best Paper Award** at the High Performance Computing and Big Data Analytics (TopHPC) Congress

2017 **2nd place** among B.Sc. Computer Engineering students at Ferdowsi University of Mashhad; directly accepted into M.Sc. program without entrance qualification exam

Skills

GenAI	LLM Applications, Agentic RAG, LangChain, MCP, In-Context Learning, Embeddings & Vector Search, Reranking
Data Eng	Milvus / Zilliz Cloud, Spark, Large-scale ETL & Data Wrangling, Entity Resolution, Data Imputation, Web/Table Data Extraction, Document Parsing & Chunking, PII Redaction
ML Stack	Supervised and Unsupervised Learning, PyTorch, Transformers / Hugging Face, Time Series Forecasting, scikit-learn, Pandas, NumPy
DevOps	CI/CD Pipelines, Git, GitHub Actions, Docker, Linux (LPIC-1), HPC/SLURM Clusters, GCP
Evaluation	RAG Evaluation (retrieval recall, accuracy, latency, cost), Benchmark Design, Ablation & Error Analysis, Regression Testing
Languages	Python, Java, SQL, Bash, MATLAB

Publications

- 2027 **EdgeLM: Edge Demonstrations for Language Models' Table Understanding**, *Soroush Omidvartehrani*, Mohammadamin Habibollah, Mohammadreza Daviran, Davood Rafiei, EACL 2027 (submitted) [paper] [code]
- 2026 **LDI: Localized Data Imputation for Text-Rich Tables**, *Soroush Omidvartehrani*, Davood Rafiei, Tabular Data Analysis (TaDA) Workshop @ VLDB 2026, Boston, USA [paper] [code]
- 2025 **WebTableX: Efficiently Discovering Web Table Transformations Through Sampling**, *Soroush Omidvartehrani*, Arash Dargahi Nobari, Davood Rafiei, Web Conference 2025 (WWW25), Sydney, Australia [paper] [code]
- 2024 **GXJoin: Generalized Cell Transformations for Explainable Joinability**, *Soroush Omidvartehrani*, Arash Dargahi Nobari, Davood Rafiei, European Conference on Advances in Databases and Information Systems (ADBIS 2024), Bayonne, France [paper] [code]
- 2023 **Discovering and Integrating Tabular Data**, Davood Rafiei, Arash Dargahi Nobari, *Soroush Omidvartehrani*, Tabular Data Analysis (TaDA) Workshop @ VLDB 2023, Vancouver, Canada [paper]
- 2022 **Online Electricity Theft Detection Framework for Large-Scale Smart Grid Data**, *Soroush Omidvartehrani*, Afshin Shahrestani, Mohammad Hossein Yaghmaee Moghaddam, Electric Power Systems Research Journal [paper]
- 2021 **Filter Based Time-Series Anomaly Detection in AMI using AI Approaches**, Alireza Rahimi, Afshin Shahrestani, Sina Ramezani, Pedram Zamani, *Soroush Omidvartehrani*, Mohammad Hossein Yaghmaee Moghaddam, 5th International Conference on Internet of Things and Applications, Isfahan, Iran [paper]
- 2021 **Anomaly in Power Consumption** (in Persian), *Soroush Omidvartehrani*, Mohammad Hossein Yaghmaee Moghaddam, ISBN: 978-622-7605-38-9, Toloo Majd, Iran
- 2020 **Decision Tree based Electricity Theft Detection in Smart Grid**, *Soroush Omidvartehrani*, Mohammad Hossein Yaghmaee Moghaddam, Mohsen Asadi, 4th International Conference on Internet of Things and Applications, Mashhad, Iran [paper]
- 2019 **Extracting Effective Features for Descriptive Analysis of Household Energy Consumption using Smart Home Data**, Hadise Moradi, *Soroush Omidvartehrani*, Behshid Behkamal, Haleh Amintoosi, High Performance Computing and Big Data Analytics Congress, Tehran, Iran [paper]
- 2019 **Analysis of Electricity Consumption Patterns in Smart Homes Using Temporal Hierarchies** (in Persian), *Soroush Omidvartehrani*, Hadise Moradi, Behshid Behkamal, Haleh Amintoosi, 3rd International Conference on Internet of Things and Applications, Isfahan, Iran
- 2018 **FUMBOT: Design, Implementation and Detection**, *Soroush Omidvartehrani*, Haleh Amintoosi, 9th OIC-CERT Annual Conference & 4th Conference on Cyberspace Security Incidents and Vulnerabilities, Shiraz, Iran

References

Dr. Davood Rafiei

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Dr. Mohammad Hossein Yaghmaee Moghaddam

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